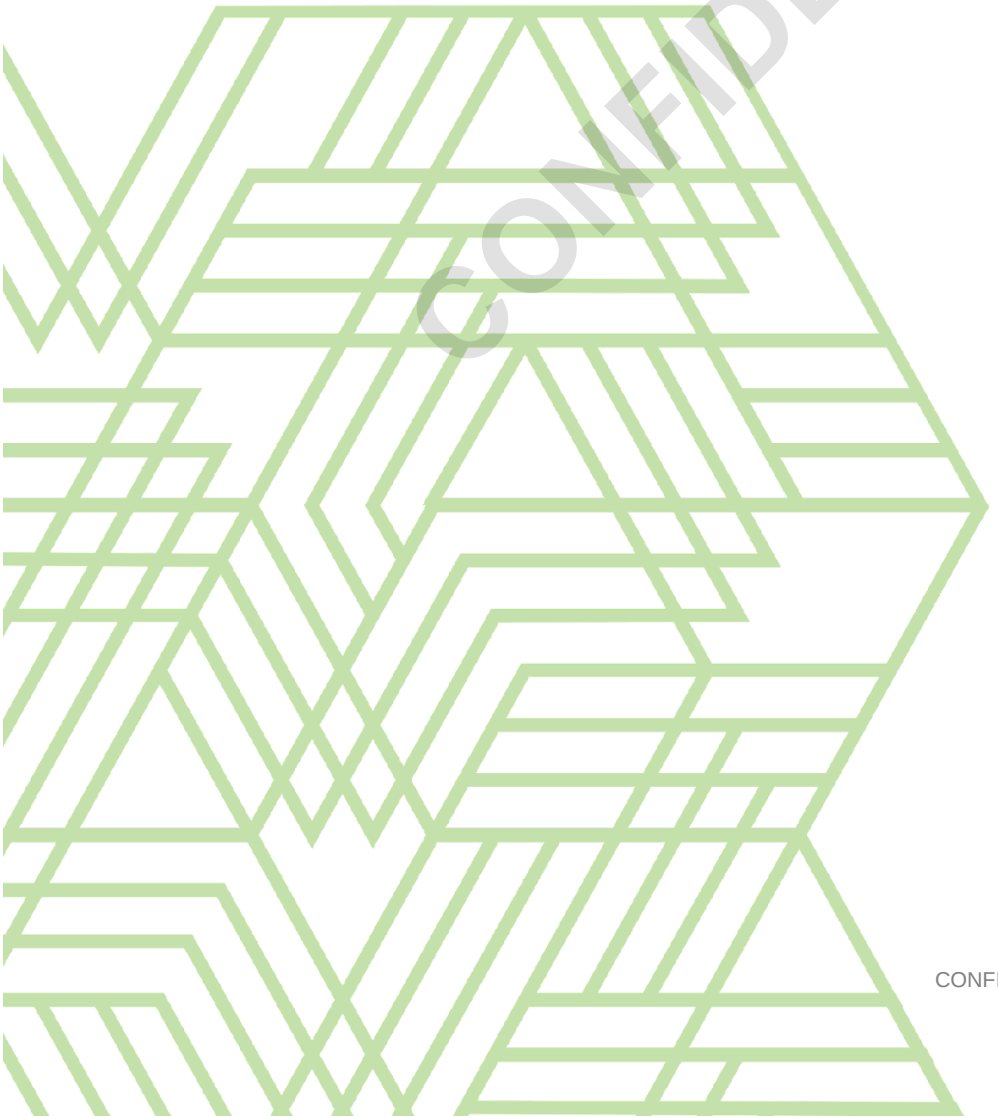
ATM33/e Series EVK

Power Consumption Evaluation

User Guide

SUMMARY: This document provides instructions for ATM33/e Evaluation Kit (EVK) users to perform power consumption evaluation of the ATM33/e Wireless SoC Series. Test setup and power consumption profiles are included in this document.

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1. Overview

This guide provides instructions for Evaluation Kit (EVK) users to perform a power consumption evaluation of the ATM33/ATM33e (a.k.a ATM33/e) Wireless SoC Series.

- Applicable Devices:
  - ATM3330e Series
    - ATM3330E-5DCAQN (Rev. B0)
  - ATM3330 Series
    - ATM3330-5DCAQN (Rev. B0)
    - ATM3325-5DCAQK (Rev. B0)
- Applicable EVKs:
  - ATMEVK-3330e-QN (for ATM3330E-5DCAQN)
  - ATMEVK-3330-QN (for ATM3330-5DCAQN)
  - ATMEVK-3325-QK (for ATM3325-5DCAQK)

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2. Evaluation Board Setup

Figure 1 shows the power consumption setup for the ATM3330e Evaluation Board (EVB).

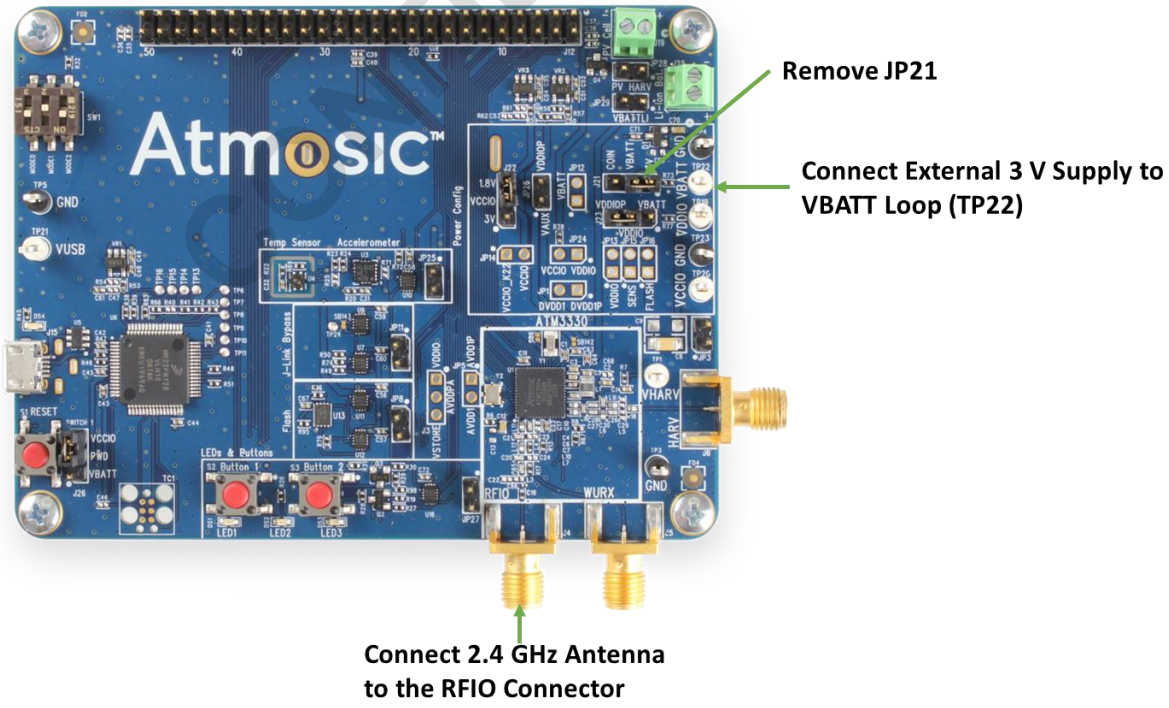


Figure 1 - ATM3330e Evaluation Board Power Consumption Setup

The setup on the other EVBs based on the ATM33 series devices are similar

1) Attach the 2.4 GHz antenna to the RFIO connector of the EVB.

2) Connect a 3 V power supply to the VBATT loop at the right side of the board, and remove the jumper on J21
3) A DC power analyzer, 6-1/2 digit multimeter, or 10 Ω resistor and oscilloscope can be used for power measurements.

2.1 EVB Configuration

The default Power Configuration jumpers are shown in Figure 2, which sets the VBATT to be 3.0 V (J21), VDDIO at 1.8 V (J23), and sets VCCIO at 1.8 V (J22) I/O voltage for the SEGGER J-Link K22 chip. The other headers are by default not populated.

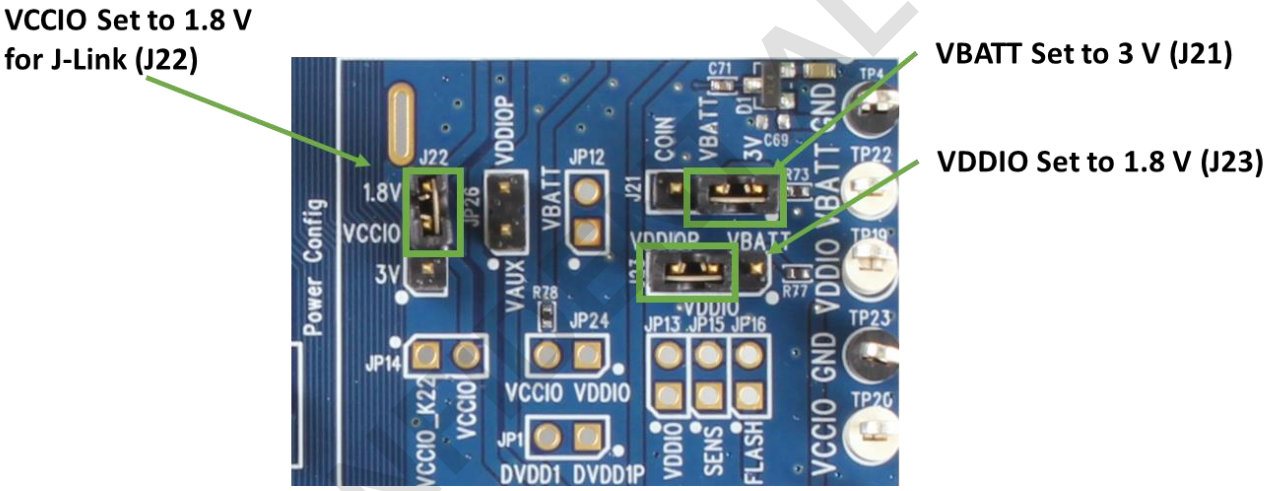


Figure 2 - ATM3330e Evaluation Board Default Configuration Jumpers

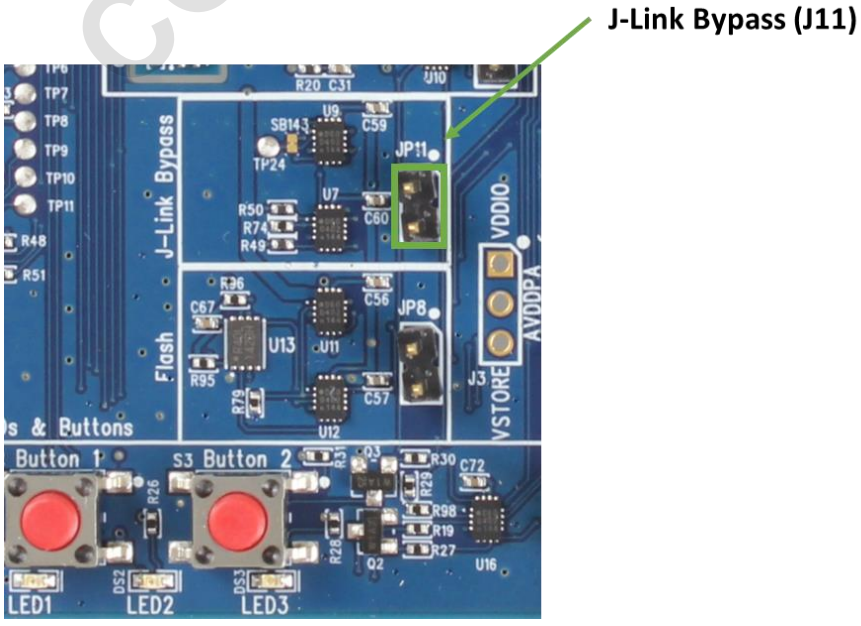


Figure 3 - EVB Power Measurement Jumpers Configuration - J-Link Bypass

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To configure the EVB for power measurements, please follow these instructions: (see Figure 2 and Figure 3 for jumper locations)

1) Remove the J21 jumper to power VBATT from the on-board 3 V regulator.
2) Connect power/ground to the VBATT/GND loops at the right edge of the EVB.
3) Power on the EVB.
4) Please use the SDK to compile and load the BLE_adv application using the power_profile reference beacon configuration. Figure 4 shows the reference beacon power profile. Details in the Configure the BLE_adv application section.
5) Please make sure the USB port to the MK22 chip remains connected at all times during the power measurement.
6) Add a jumper on JP11 to disconnect the MK22 and prevent it from drawing power from or providing power to the ATM33 device (See Figure 3). Remove this jumper once the power measurement is done.

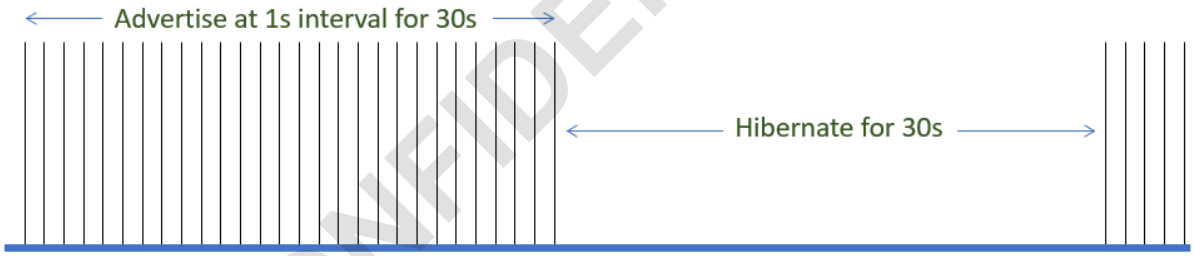


Figure 4 - Reference Beacon Power Profile

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2.2 Configure the BLE_adv Application

First, please add following flags in makefile to configure the switching regulator optimally for the default EVB power configuration.

```
CFLAGS += \
          -DBATT_LEVEL=BATT_LEVEL_GT_1P8V \
          -DVDDIO_TYPE=VDDIO_TYPE_INTERNAL
```

Next, compile and load the BLE_adv application to use the power profile configuration. Example command:

```
make run_all REF_BCN=power_profile
```

Please see the **Atmosic SDK Quick Start Guide** for more information on loading a beacon application using the SDK.

3. Power Measurement Procedures

3.1 Average Power Measurement with Multimeter

1) Connect a 6½ digit multimeter in series with the 3 V supply.
2) Set the multimeter for DC Current and increase averaging to as long as possible. For example, the Keysight 34465A averaging time can be set to 1s.

3.2 Average Power Profile Measurement

1) Connect a 10 Ω resistor in series between the ground of the bench power supply with the ground of the EVB.
2) Put an oscilloscope probe across the 10 Ω resistor.



Figure 5 - Approximate Power Profile Measurement

3) The oscilloscope will show a periodic beacon profile (3 pulses for 3 channels) during the advertising phase followed by the hibernation phase without beacons. See Figure 5.
4) The approximate maximum transmit power can be approximated by dividing the measured peak voltage across the 10 Ω resistor. It should be approximately 2.5 mA. Note that the oscilloscope will not provide enough resolution to measure other power states.

3.3 More Accurate Power Profile Measurement

Measuring the dynamic current more accurately requires a DC power analyzer such as the Keysight N6705C. For the Keysight N6705C, it is important to use auto-ranging and the maximum number of horizontal data points to observe the most accurate power profile. Figures 6 to Figure 10 show current consumption measurements of various states.

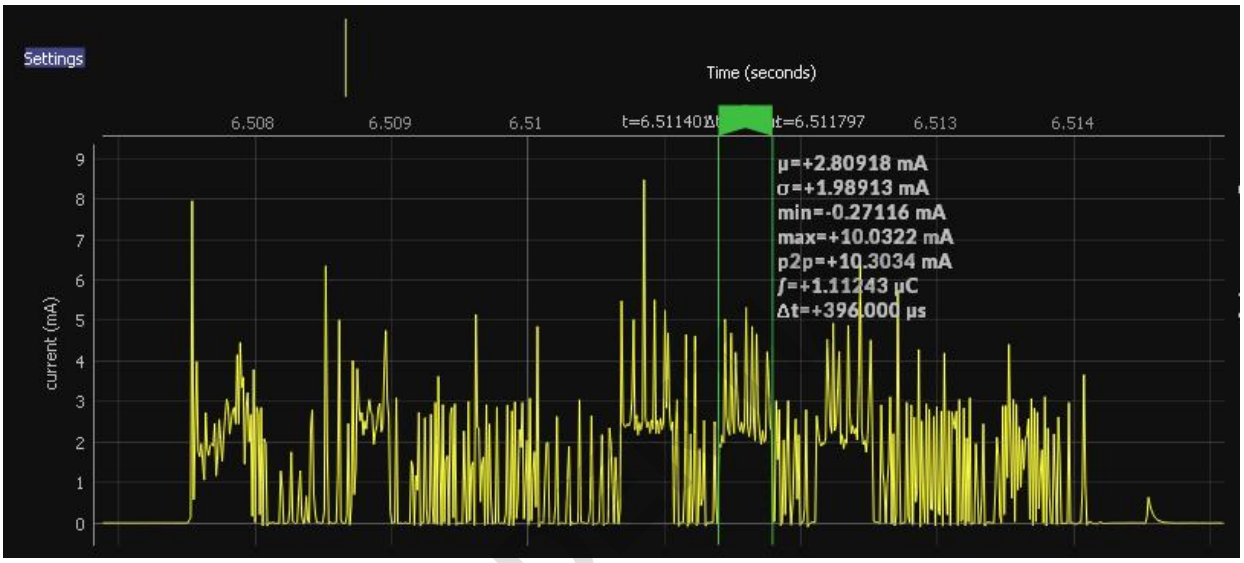


Figure 6 - Transmit Current Profile

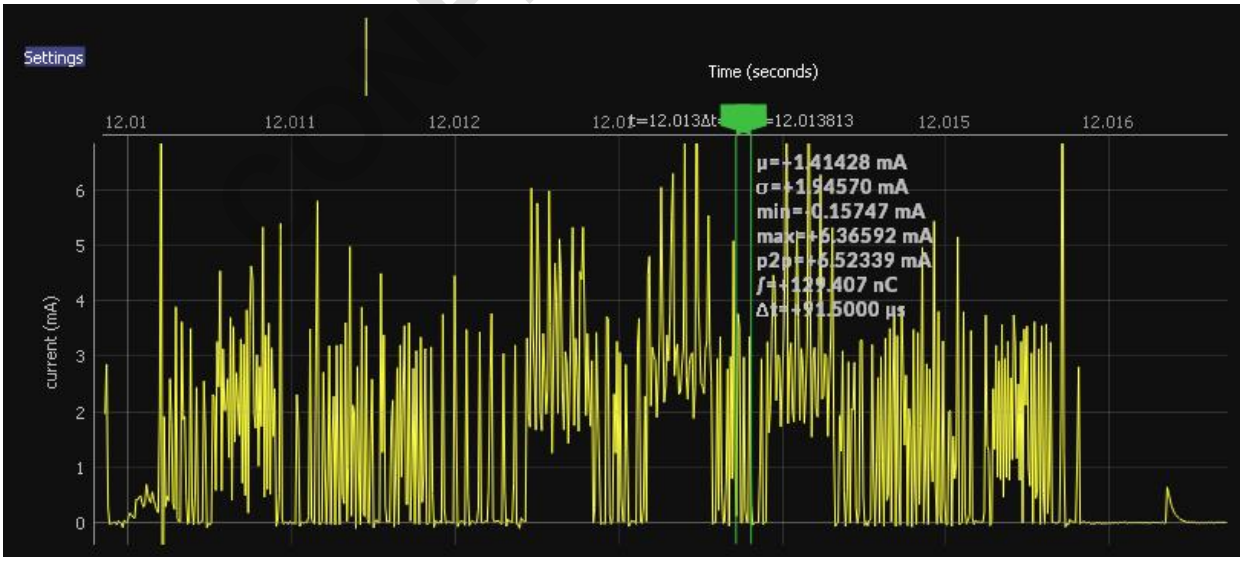


Figure 7 - Receive Current Profile

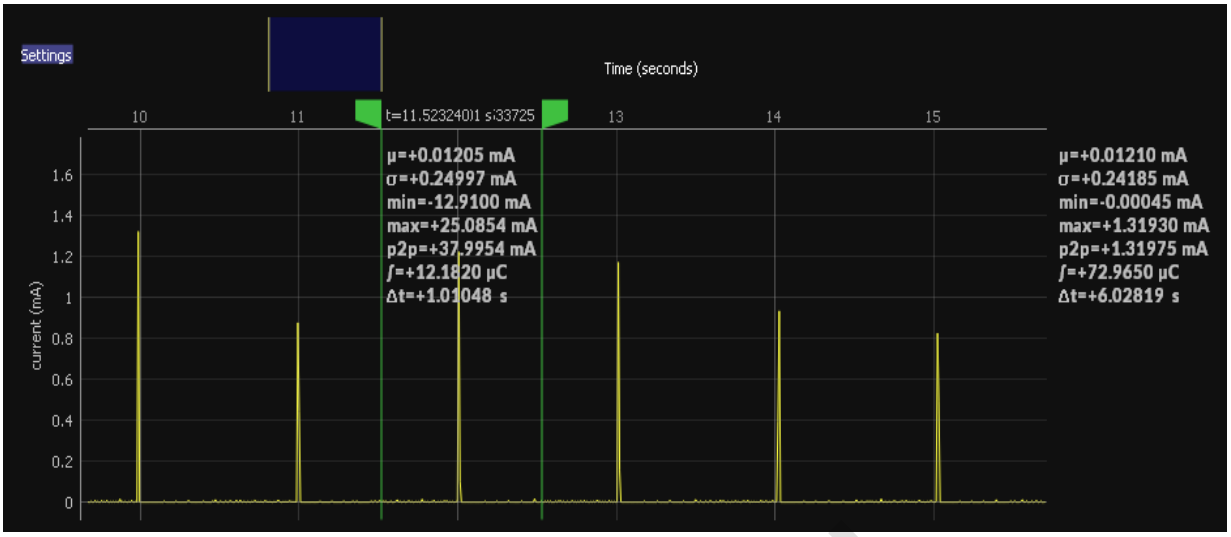


Figure 8 - Current Profile Over 1s Interval During Advertising Phase

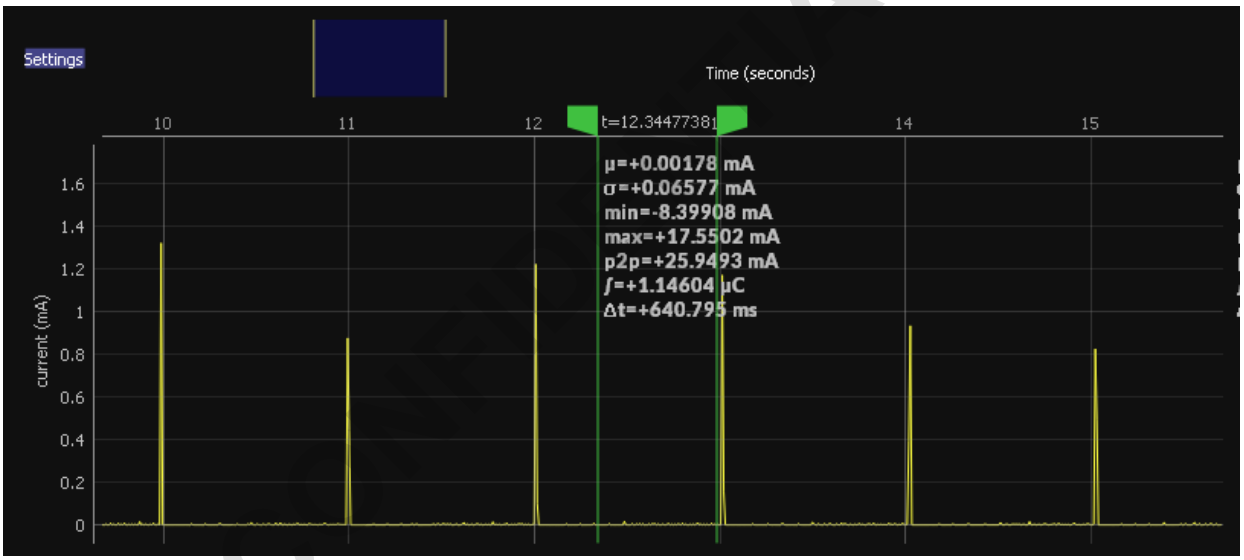


Figure 9 - Retention Current Profile

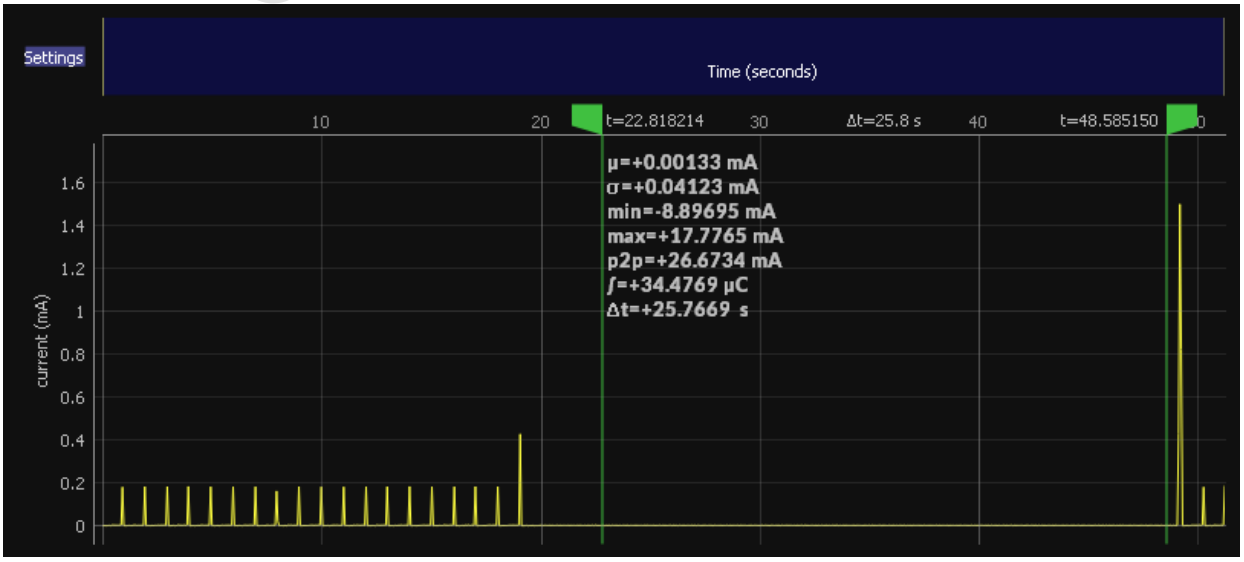


Figure 10 - Hibernation Current Profile

Table 1 summarizes typical EVB current measurements for each of the above states.

Table 1 - Typical Current for Various States for ATM33/ATM33e Wireless SoC Series

State	Average current on VBAT = 3 V (µA)
Transmit (0 dBm)	2500
Receive	1400
1s interval during advertising phase	12.1
Retention	1.8
Hibernation	1.3

It is normal to observe peaks during low power or active modes such as the ones in Figure 9 and Figure 10. These peaks result from typical operation of the DC/DC switching regulator and are generally harmless in most applications.

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Revision History

Date	Version	Description
August 31, 2022	0.51	Updated Figure 1 - ATM3330e Evaluation Board Power Consumption Setup, Figure 2 - ATM3330e Evaluation Board Default Configuration Jumpers, Figure 3 - EVB Power Measurement Jumpers Configuration - J-Link Bypass, Figure 6 - Transmit Current Profile, Figure 7 - Receive Current Profile, Figure 8 - Current Profile Over 1s Interval During Advertising Phase,Figure 9 - Retention Current Profile, Figure 10 - Hibernation Current Profile Table 1 - Typical Current for Various States for ATM33/ATM33e Wireless SoC Series,
May 13, 2022	0.50	Initial version created

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